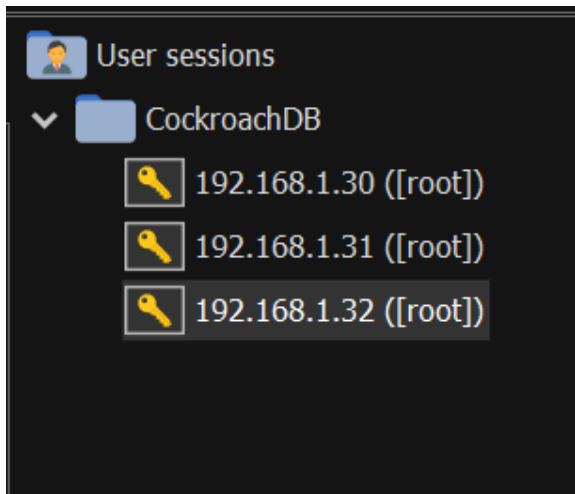


Multi-Region Availability for OLTP Database

Yunata Gunawan – January 2026



Pre-Requisite



This is only for PoC purposes. Currently I have 3 VMs.

I'm going to simulate 3 cluster with each cluster contains 3 instances (simulate 3 VMs).

I'm also set up a load balancer to distribute the traffic.

The actual production must use more than 3 VMs + additional VMs for load balancer.

AWS Regions and Availability Zones

AWS Regions and Availability Zones < User Guide		ap-southeast-1	Asia Pacific (Singapore)	3	Singapore	Not required
		ap-southeast-2	Asia Pacific (Sydney)	3	Australia	Not required
AWS Regions and Availability Zones < User Guide		ap-southeast-3	Asia Pacific (Jakarta)	3	Indonesia	Required
		ap-southeast-5	Asia Pacific (Malaysia)	3	Malaysia	Required

<https://docs.aws.amazon.com/global-infrastructure/latest/regions/aws-regions.html>

Software Installation [All VMs]

CPU Architecture & OS Version

```
[root@second cockroachdb]# lscpu
cat /etc/os-release
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 4
On-line CPU(s) list:    0-3
```

```
[root@second cockroachdb]# cat /etc/os-release
NAME="Rocky Linux"
VERSION="9.7 (Blue Onyx)"
ID="rocky"
ID_LIKE="rhel centos fedora"
VERSION_ID="9.7"
PLATFORM_ID="platform:el9"
PRETTY_NAME="Rocky Linux 9.7 (Blue Onyx)"

```

Download Binary

wget

<https://binaries.cockroachdb.com/cockroach-v25.3.6.linux-amd64.tgz>

tar -xvzf cockroach-v25.3.6.linux-amd64.tgz

```
[root@second ~]# ls -altr
total 135316
-rw-r--r-- 1 root root 129 May 4 2025 .tcshrc
-rw-r--r-- 1 root root 100 May 4 2025 .cshrc
-rw-r--r-- 1 root root 429 May 4 2025 .bashrc
-rw-r--r-- 1 root root 141 May 4 2025 .bash_profile
-rw-r--r-- 1 root root 18 May 4 2025 .bash_logout
drwxr-xr-x 18 root root 235 Oct 24 22:18 ..
-rw-r----- 1 root root 887 Oct 24 22:36 anaconda-ks.cfg
-rw-r--r-- 1 root root 13849412 Dec 18 03:03 cockroach-v25.3.6.linux-amd64.tgz
drwx----- 2 root root 6 Jan 1 22:15 .ssh
-rw-r----- 1 root root 532 Jan 1 22:33 .viminfo
-rw-r----- 1 root root 20 Jan 2 21:44 .lessht
drwxr-x--- 5 root root 4096 Jan 2 22:49 ap-southeast-2a
drwxr-x--- 5 root root 4096 Jan 2 22:49 ap-southeast-2b
drwxr-x--- 5 root root 4096 Jan 2 22:51 ap-southeast-2c
-rw-r--r-- 1 root root 229 Jan 2 23:35 .cockroachsql_history
dr-xr-x--- 7 root root 4096 Jan 2 23:35 .
-rw-r----- 1 root root 9725 Jan 3 02:51 .bash_history
drwxr-xr-x 8 root root 187 Jan 16 10:00 cockroachdb
[root@second ~]#
```

Network Time Protocol [All VMs]

```
[root@third ~]# cat /etc/chrony.conf
# Use public servers from the pool.ntp.org
# Please consider joining the pool (http://www.pool.ntp.org/join.html)

server time1.google.com iburst
server time2.google.com iburst
server time3.google.com iburst
server time4.google.com iburst
```

```
[root@first ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^+ time1.google.com         1 10 377 762 -179us[ -177us] +/- 801us
^+ time2.google.com         1 10 377 770 +199us[ +200us] +/- 1035us
^- time3.google.com         1 10 377 856 -64us[ -62us] +/- 20ms
^* time4.google.com         1 10 377 636 +68us[ +69us] +/- 669us
```

```
[root@second ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^+ time1.google.com         1 10 377 476 +134us[ +134us] +/- 815us
^+ time2.google.com         1 10 377 753 +40us[ +40us] +/- 1094us
^- time3.google.com         1 10 377 613 +193us[ +193us] +/- 20ms
^* time4.google.com         1 10 377 981 +182us[ +179us] +/- 798us
```

```
[root@third ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* time1.google.com         1 10 377 569 -37us[ -38us] +/- 996us
^+ time2.google.com         1 10 377 104 -61us[ -61us] +/- 829us
^- time3.google.com         1 10 377 549 +244us[ +244us] +/- 20ms
^+ time4.google.com         1 10 377 62 +59us[ +59us] +/- 951us
```

```
restrict 192.168.1.0 netmask 255.255.255.0
nomodify notrap
sudo firewall-cmd --add-service=ntp --permanent
sudo firewall-cmd --reload
sudo systemctl restart chronyd
sudo systemctl enable chronyd
sudo systemctl status chronyd
chronyc sources
```

Self Signed Certificate

```
# Security Certificate
```

```
mkdir certs
```

```
mkdir my-safe-directory
```

```
./cockroach cert create-ca --certs-dir=certs --ca-key=my-safe-directory/ca.key
```

```
./cockroach cert create-client root --certs-dir=/root/cockroachdb/certs --ca-key=my-safe-directory/ca.key --overwrite
```

```
scp my-safe-directory/ca.key root@192.168.1.31:/root/cockroachdb/my-safe-directory
```

```
scp my-safe-directory/ca.key root@192.168.1.32:/root/cockroachdb/my-safe-directory
```

```
scp certs/ca.crt certs/client.root.crt certs/client.root.key root@192.168.1.31:/root/cockroachdb/certs
```

```
scp certs/ca.crt certs/client.root.crt certs/client.root.key root@192.168.1.32:/root/cockroachdb/certs
```

```
./cockroach cert create-node 192.168.1.30 192.168.1.30 first localhost 127.0.0.1 --certs-dir=certs --ca-key=my-safe-directory/ca.key --overwrite
```

```
./cockroach cert create-node 192.168.1.31 192.168.1.31 second localhost 127.0.0.1 --certs-dir=certs --ca-key=my-safe-directory/ca.key --overwrite
```

```
./cockroach cert create-node 192.168.1.32 192.168.1.32 third localhost 127.0.0.1 --certs-dir=certs --ca-key=my-safe-directory/ca.key --overwrite
```

```
# Make Interface is Trusted to access on all port
```

```
sudo firewall-cmd --zone=trusted --change-interface=ens18 --permanent
```

```
sudo firewall-cmd --reload
```

```
firewall-cmd --zone=trusted --list-all
```

Start 3 Instance on First VM

```
# 192.168.1.30 (192.168.1.30)
./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.30:26257 \
--join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259 \
--http-addr=192.168.1.30:8080 --locality=region=ap-southeast-1,zone=ap-southeast-1a --store=ap-southeast-1a --max-offset
250ms --background
```

```
./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.30:26258 \
--join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259 \
--http-addr=192.168.1.30:8081 --locality=region=ap-southeast-1,zone=ap-southeast-1b --store=ap-southeast-1b --max-offset
250ms --background
```

```
./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.30:26259 \
--join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259 \
--http-addr=192.168.1.30:8082 --locality=region=ap-southeast-1,zone=ap-southeast-1c --store=ap-southeast-1c --max-offset
250ms --background
```

Execute only once

```
./cockroach init --certs-dir=/root/cockroachdb/certs --host 192.168.1.30:26258
```

Start 3 Instance on Second VM

```
# 192.168.1.31 (192.168.1.31)
./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.31:26257 \
--join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259,192.168.1.31:26257,192.168.1.31:26258,192.168.1.31:26259 \
--http-addr=192.168.1.31:8080 --locality=region=ap-southeast-2,zone=ap-southeast-2a --store=ap-southeast-2a --max-offset
250ms --background
```

```
./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.31:26258 \
--join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259,192.168.1.31:26257,192.168.1.31:26258,192.168.1.31:26259 \
--http-addr=192.168.1.31:8081 --locality=region=ap-southeast-2,zone=ap-southeast-2b --store=ap-southeast-2b --max-offset
250ms --background
```

```
./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.31:26259 \
--join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259,192.168.1.31:26257,192.168.1.31:26258,192.168.1.31:26259 \
--http-addr=192.168.1.31:8082 --locality=region=ap-southeast-2,zone=ap-southeast-2c --store=ap-southeast-2c --max-offset
250ms --background
```

Start 3 Instance on Third VM

```
# 192.168.1.32 (192.168.1.32)
./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.32:26257 \
--
join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259,192.168.1.31:26257,192.168.1.31:26258,192.168.1.31:26259,192.168.1.32:26257,192.168.1.32:26258,192.168.1.32:26259 \
--http-addr=192.168.1.32:8080 --locality=region=ap-southeast-3,zone=ap-southeast-3a --store=ap-southeast-3a --max-offset 250ms --background

./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.32:26258 \
--
join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259,192.168.1.31:26257,192.168.1.31:26258,192.168.1.31:26259,192.168.1.32:26257,192.168.1.32:26258,192.168.1.32:26259 \
--http-addr=192.168.1.32:8081 --locality=region=ap-southeast-3,zone=ap-southeast-3b --store=ap-southeast-3b --max-offset 250ms --background

./cockroach start --certs-dir=/root/cockroachdb/certs --listen-addr=192.168.1.32:26259 \
--
join=192.168.1.30:26257,192.168.1.30:26258,192.168.1.30:26259,192.168.1.31:26257,192.168.1.31:26258,192.168.1.31:26259,192.168.1.32:26257,192.168.1.32:26258,192.168.1.32:26259 \
--http-addr=192.168.1.32:8082 --locality=region=ap-southeast-3,zone=ap-southeast-3c --store=ap-southeast-3c --max-offset 250ms --background
```

Create Admin User

```
Cluster successfully initialized
[root@first cockroachdb]# ./cockroach sql --host 192.168.1.30 --port 26257 --user root --certs-dir=/root/cockroachdb/certs
#
# Welcome to the CockroachDB SQL shell.
# All statements must be terminated by a semicolon.
# To exit, type: \q.
#
# Server version: CockroachDB CCL v25.3.6 (x86_64-pc-linux-gnu, built 2025/12/16 12:08:06, go1.23.12 X:nocoverageredesign) (same version as client)
# Cluster ID: 3221c3bb-2d53-420f-b6a3-88aeaecf54e1
#
# Enter \? for a brief introduction.
#
```

```
./cockroach sql --host 192.168.1.30 --port 26257 --user root --certs-dir=/root/cockroachdb/certs
select node_id, address, locality, started_at from crdb_internal.kv_node_status;
#ALTER USER root WITH PASSWORD 'root';
CREATE USER adminoltp WITH LOGIN PASSWORD 'P@ssw0rd' VALID UNTIL '9999-12-31';
GRANT admin TO adminoltp;
```

CockroachDB Web UI (Any Nodes Can Do)

← → ↺

Not secure https://192.168.1.30:8080/#/overview/list

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⋮

 CockroachDB

A

Cluster id: 3221c3bb-2d53-420f-b6a3-88aeaecf54e1

V25.3.6

This cluster will require a license key by January 23rd, 2026 or the cluster will be throttled. [Learn more](#)

Overview

Metrics

Databases

SQL Activity

Insights

Network

Hot Ranges

Jobs

Schedules

Advanced Debug

Capacity Usage

0.3%

0 GiB

200 GiB

400 GiB

USED

1.5 GiB

USABLE

547.6 GiB

Node Status

9

0

0

0

56

0

0

LIVE
NODES

SUSPECT
NODES

DRAINING
NODES

DEAD
NODES

TOTAL
NODERANGES

UNDER-REPLICATED
RANGES

UNAVAILABLE
RANGES

Replication Status

0

56

0

0

Node List ▾

Nodes (9)

nodes	Node Count	Uptime	Replicas	Capacity Usage	Memory Use	vCPUs	Version	Status
-------	---------------	--------	----------	-------------------	---------------	-------	---------	--------

Stop And Delete All Data

[Execute on All VMs]

```
pkill -9 cockroach
```

```
rm -rf ap-southeast-*
```

Creating Multi Region Database and Table

```
SHOW DATABASES;
SHOW REGIONS;
SHOW REGIONS FROM CLUSTER;
SHOW REGIONS FROM DATABASE <DBNAME>;
CREATE DATABASE multiregion;
ALTER DATABASE multiregion SET PRIMARY REGION 'ap-southeast-1';
ALTER DATABASE multiregion ADD REGION 'ap-southeast-2';
ALTER DATABASE multiregion ADD REGION 'ap-southeast-3';
ALTER DATABASE multiregion SET SECONDARY REGION 'ap-southeast-2';
SHOW ALL ZONE CONFIGURATIONS;
SHOW ZONE CONFIGURATION FROM DATABASE multiregion;
ALTER DATABASE multiregion SURVIVE REGION FAILURE;
SHOW REGIONS FROM DATABASE multiregion;
USE multiregion;
CREATE TABLE poc_test (id INT PRIMARY KEY NOT NULL, name VARCHAR(25)
NULL, dob DATE NULL);
INSERT INTO poc_test VALUES (1, 'siska', CURRENT_TIMESTAMP);
INSERT INTO poc_test VALUES (2, 'john', CURRENT_TIMESTAMP);
SELECT * FROM poc_test;
SHOW CREATE TABLE poc_test;
SHOW RANGES FROM TABLE poc_test WITH DETAILS;
```

```
root@192.168.1.30:26257/defaultdb> SHOW ZONE CONFIGURATION FROM DATABASE multiregion;
+-----+-----+
| target | raw_config_sql |
+-----+-----+
| DATABASE multiregion | ALTER DATABASE multiregion CONFIGURE ZONE USING
| range_min_bytes = 134217728,
| range_max_bytes = 536870912,
| gc_ttlseconds = 14400,
| num_replicas = 5,
| num_voters = 3,
| constraints = '{+region=ap-southeast-1: 1, +region=ap-southeast-2: 1, +region=ap-southeast-3: 1}'
|
| voter_constraints = '[+region=ap-southeast-1]',
| lease_preferences = '[[+region=ap-southeast-1], [+region=ap-southeast-2]]'
|
+-----+-----+
(1 row)
```

```
root@192.168.1.30:26257/defaultdb> SHOW ZONE CONFIGURATION FROM DATABASE multiregion;
+-----+-----+
| target | raw_config_sql |
+-----+-----+
| DATABASE multiregion | ALTER DATABASE multiregion CONFIGURE ZONE USING
| range_min_bytes = 134217728,
| range_max_bytes = 536870912,
| gc_ttlseconds = 14400,
| num_replicas = 5,
| num_voters = 5,
| constraints = '{+region=ap-southeast-1: 1, +region=ap-southeast-2: 1, +region=ap-southeast-3: 1}'
|
| voter_constraints = '{+region=ap-southeast-1: 2, +region=ap-southeast-2: 2}',
| lease_preferences = '[[+region=ap-southeast-1], [+region=ap-southeast-2]]'
|
+-----+-----+
(1 row)
```

Time: 3ms total (execution 3ms / network 0ms)

```
root@192.168.1.30:26257/defaultdb> SHOW REGIONS FROM DATABASE multiregion;
+-----+-----+-----+-----+-----+
| database | region | primary | secondary | zones |
+-----+-----+-----+-----+-----+
| multiregion | ap-southeast-1 | t | f | {ap-southeast-1a,ap-southeast-1b,ap-southeast-1c}
| multiregion | ap-southeast-2 | f | t | {ap-southeast-2a,ap-southeast-2b,ap-southeast-2c}
| multiregion | ap-southeast-3 | f | f | {ap-southeast-3a,ap-southeast-3b,ap-southeast-3c}
+-----+-----+-----+-----+-----+
(3 rows)
```

```
root@192.168.1.30:26257/multiregion> SHOW CREATE TABLE poc_test;
+-----+-----+
| table_name | create_statement |
+-----+-----+
| poc_test | CREATE TABLE public.poc_test (
| id INT8 NOT NULL,
| name VARCHAR(25) NULL,
| dob DATE NULL,
| CONSTRAINT poc_test_pkey PRIMARY KEY (id ASC)
| ) LOCALITY REGIONAL BY TABLE IN PRIMARY REGION;
+-----+-----+
(1 row)
```

Time: 17ms total (execution 17ms / network 0ms)

```
root@192.168.1.30:26257/multiregion> SHOW REGIONS;
+-----+-----+-----+-----+-----+
| region | zones | database_names | primary_region_of | secondary_region_of |
+-----+-----+-----+-----+-----+
| ap-southeast-1 | {ap-southeast-1a,ap-southeast-1b,ap-southeast-1c} | {multiregion} | {multiregion} | {}
| ap-southeast-2 | {ap-southeast-2a,ap-southeast-2b,ap-southeast-2c} | {multiregion} | {} | {multiregion}
| ap-southeast-3 | {ap-southeast-3a,ap-southeast-3b,ap-southeast-3c} | {multiregion} | {} | {}
+-----+-----+-----+-----+-----+
(3 rows)
```

Time: 19ms total (execution 19ms / network 0ms)

Web UI Database Regions Replication

← → ↻

Not secure https://192.168.1.30:8080/#/databases?page=1&pageSize=10&sortField=name&sortOrder=asc

☆

Y

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 CockroachDB

A

Cluster id: 3221c3bb-2d53-420f-b6a3-88aeaecf54e1

V25.3.6

Your license key expired on January 16th, 2026. The cluster will be throttled on January 23rd, 2026 unless a new license key is added. [Learn more](#)

Overview

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Databases

Auto stats collection - enabled

Search databases

Select Nodes

1-4 of 4 databases

Last refreshed: a few seconds ago

Name	Size	Tables	Regions / Nodes
defaultdb	0 B	0	
multiregion	48.2 KiB	1	ap-southeast-1 2 ap-southeast-2 2 ap-southeast-3 1

Check Table Replication

```
root@192.168.1.30:26257/multiregion> INSERT INTO poc_test VALUES (1, 'siska', CURRENT_TIMESTAMP);
INSERT 0 1

Time: 24ms total (execution 24ms / network 0ms)

root@192.168.1.30:26257/multiregion> INSERT INTO poc_test VALUES (2, 'john', CURRENT_TIMESTAMP);
INSERT 0 1

Time: 16ms total (execution 16ms / network 0ms)

root@192.168.1.30:26257/multiregion>
^C
Use \q or terminate input to exit.
root@192.168.1.30:26257/multiregion> SELECT * FROM poc_test;
 id | name | dob
-----+-----+-----
  1 | siska | 2026-01-16
  2 | john  | 2026-01-16
(2 rows)
```

[illegible]

On the Web UI you can see on the database table section the table information..

Name	Replication Size	Ranges	Columns	Indexes	Regions / Nodes	% of Live data	Table auto stats enabled	Stats last updated
public.poc_test	48.2 KiB	1	3	1	ap-southeast-1	100.0 % 65.0 B live data / 65.0 B total	ENABLED	Jan 16, 2026 at 3:25:52 UTC
					ap-southeast-2			
					ap-southeast-3			

Web UI Network & Metric Summary

Network

Sort By: Region ▾

Filter: ▾

☒ Collapse Nodes

STANDARD DEVIATION

0.28ms

0.41ms

0.54ms

0.66ms

0.79ms

-2 std dev

-1 std dev

Mean

+1 std dev

+2 std dev

ap-southeast-1

ap-southeast-1

ap-southeast-2

ap-southeast-3

0.80ms

0.79ms

ap-southeast-2

0.64ms

0.87ms

Summary

Total Nodes [View nodes list](#) 9

Capacity Usage 0.28%

You are using 1.5 GiB of 547.6 GiB usable disk capacity across all nodes.

Unavailable ranges 0

Queries per second 0.0

Sum of Selects, Updates, Inserts, and Deletes across your entire cluster.

P99 latency 0.0 ms

Adding Load Balancer (HAProxy)

```
./cockroach gen haproxy --host 192.168.1.32 --  
certs-dir=/root/cockroachdb/certs
```

```
sudo yum install haproxy
```

```
nano ./haproxy.cfg # change bind port from  
cockroachdb port to anything
```

```
haproxy -f ./haproxy.cfg -D
```

```
[root@third cockroachdb]# haproxy -f ./haproxy.cfg -D  
[root@third cockroachdb]# ps
```

```
[root@third cockroachdb]# cat haproxy.cfg  
global  
    maxconn 4096  
defaults  
    mode                tcp  
    retries              2  
    timeout connect      5s  
    timeout client       10m  
    timeout server       10m  
    option               clitcpka  
listen psql  
    bind :26260  
    mode tcp  
    balance roundrobin  
    option httpchk GET /health?ready=1  
    server cockroach1 192.168.1.30:26258 check port 8081  
    server cockroach2 192.168.1.30:26257 check port 8080  
    server cockroach3 192.168.1.30:26259 check port 8082  
    server cockroach4 192.168.1.31:26257 check port 8080  
    server cockroach5 192.168.1.31:26258 check port 8081  
    server cockroach6 192.168.1.31:26259 check port 8082  
    server cockroach7 192.168.1.32:26257 check port 8080  
    server cockroach8 192.168.1.32:26258 check port 8081  
    server cockroach9 192.168.1.32:26259 check port 8082
```

Dbeaver JDBC Connect Test HAProxy

